

SMART INDIA HACKATHON 2026

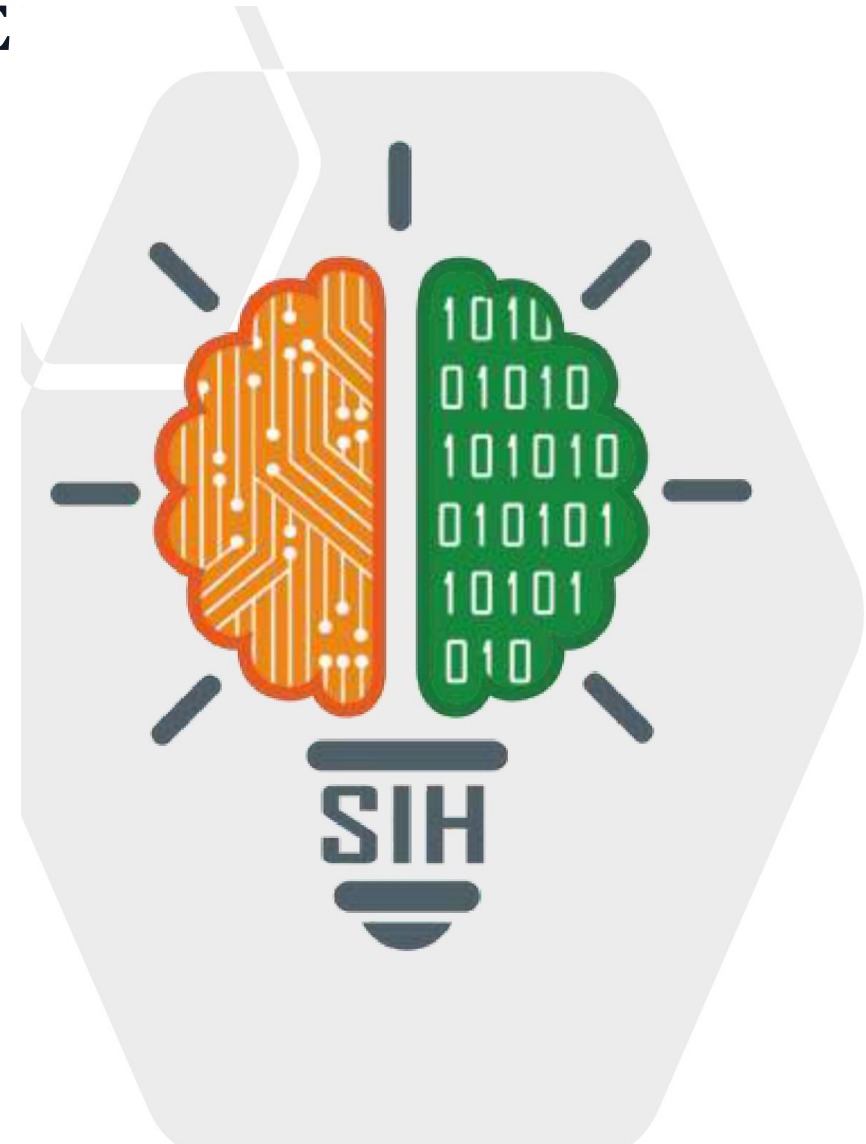


TITLE PAGE

- **Problem Statement ID – SIH26188**
- **Problem Statement Title-** AI-Based Fake Identity & Document Screening System
- **Theme-** Blockchain & Cybersecurity
- **PS Category- Software/Hardware** Software
- **Team ID- SIH-S-B1-148**
- **Team Name (Registered on portal) Asterix**

Project Name: DocuShield AI

AI-Based Fake Identity & Multimodal Document Forensic Screening System



• **Proposed Solution (Describe your Idea/Solution/Prototype)**

• **Detailed explanation of the proposed solution:**

- DocuShield AI is an automated, multimodal forensic screening platform built to verify document authenticity across 5 core identity categories: Passports, Visas, National ID (Aadhaar/PAN), Driving Licences, and Permits.
- Operates a synchronized Dual-Engine Architecture: Engine A validates text, dates, and mathematical checksums (ICAO Doc 9303 MRZ, Verhoeff D5 algorithm), while Engine B performs digital image forensics (JPEG Error Level Analysis, Copy-Move cloning detection, Typography sharpness, and EXIF software signatures).

• **How it addresses the problem:**

- **The Core Vulnerability:** Forgers easily produce cards with plausible text and valid formats, but edit photos, dates, or stamps using digital manipulation software.
- **Traditional Blindspot:** Basic OCR and rule validators only read text strings; they are blind to pixel cut seams, cloned seals, or recompression discrepancies.
- **The DocuShield Solution:** Fuses optical text semantics with pixel-level physics. If an ID number looks plausible but the photo boundary shows cut seams or ELA anomalies, the document is flagged.

• **Innovation and uniqueness of the solution:**

- ✓ **Multi-Family Evidence Fusion:** Hierarchically corroborates 3 independent evidence families (Compression, Structural/Visual, Content/Identity) into a calibrated risk score [0–100%].
- ✓ **“Why This Verdict?” Explainability:** Generates human-readable evidence items (positive checks, cautions, critical indicators) to empower human border/KYC officers rather than opaque black-box AI.
- ✓ **Deterministic & Privacy-Preserving:** Strictly local execution without requiring citizen PII storage or external government database simulation, operating securely offline or on-premise.

Calibrated Forensic Verdicts: AUTHENTIC (≥80% | Risk ≤20%) • SUSPICIOUS (50–79% | Review) • FLAGGED (<50% | Tampered)

Dedicated Image Forensics Suite: ELA (90% JPEG) | Copy-Move (ORB+RANSAC) | Typography Stroke Variance | EXIF Software Markers

• **Technologies to be used (programming languages, frameworks, hardware):**

✓ **Core Language & Backend:** Python 3.11+, FastAPI (REST API), Uvicorn ASGI server

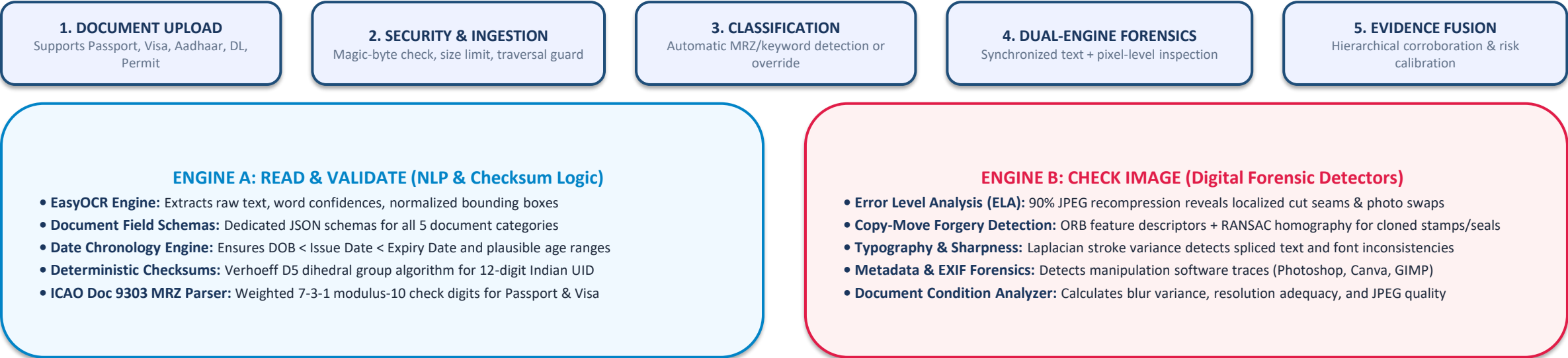
✓ **OCR & Text Parsing:** EasyOCR (Neural Network Text Recognizer), PyTorch, Tesseract fallback

✓ **Computer Vision & Forensics:** OpenCV (cv2), NumPy, Pillow (PIL), SciPy for Laplacian/ORB/ELA

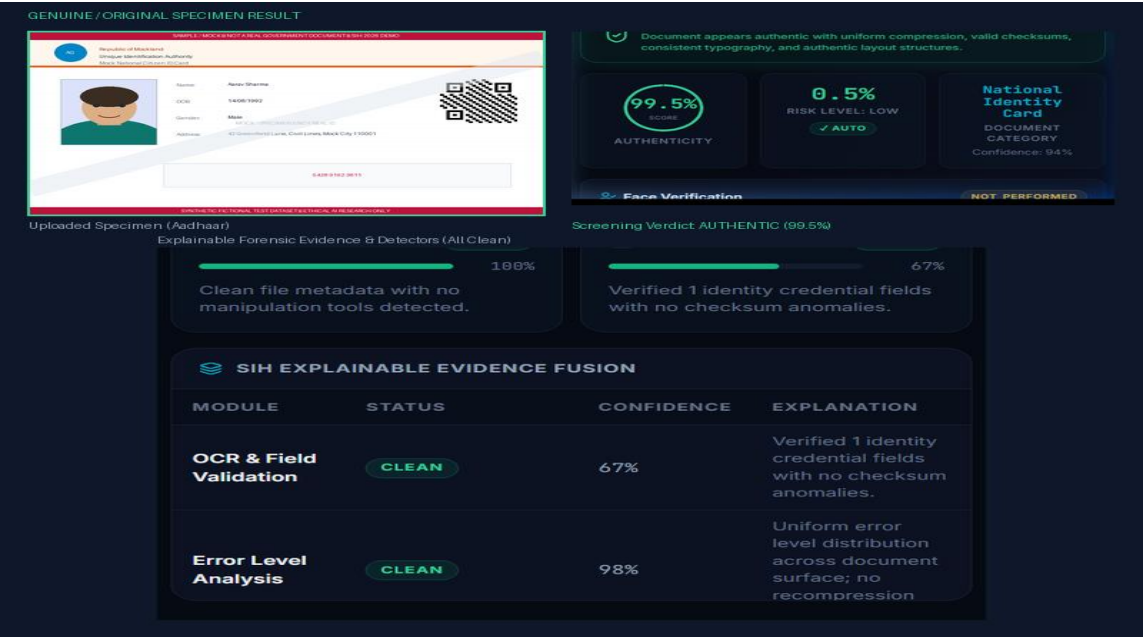
✓ **Frontend & UI Dashboard:** React 18, Vite, Canvas SVG Inspector, Lucide Icons

✓ **Hardware & Deployment:** Lightweight CPU (2 cores, 2GB RAM; optional GPU for high batch throughput)

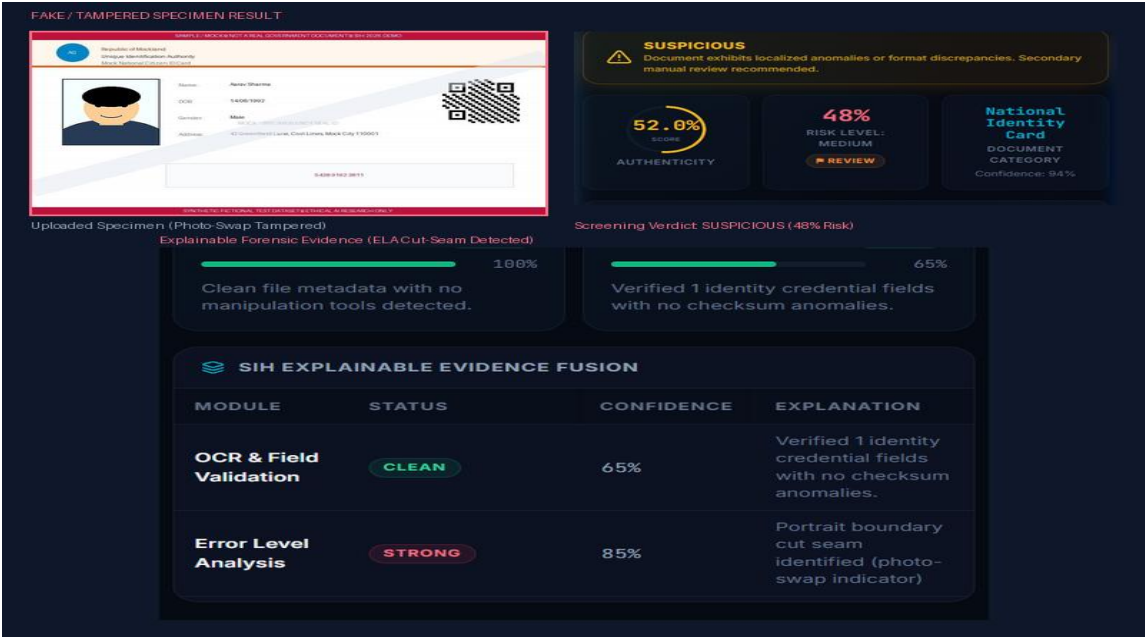
• **Methodology and process for implementation (Flow Charts / Images / working prototype):**



• Analysis of the feasibility of the idea (Validated via Live Working Prototype):



Genuine Specimen: AUTHENTIC (Score: 99.5% | Risk: 0.5% | Clean ELA & Verhoeff Valid)



Tampered Specimen: SUSPICIOUS (Score: 52.0% | Risk: 48.0% | ELA Strong 85% Cut-Seam Flagged)

• Potential challenges and risks:

- **Physical Print-and-Scan Replays:** High-resolution printed & rescanned cards can mask JPEG compression discrepancies.
- **Camera Glare & Shadows:** Extreme reflections on laminated plastic cards degrade OCR token clarity.
- **Absence of Government DB:** Fabricated identities with completely plausible details cannot be validated without central registry access.

• Strategies for overcoming these challenges:

- ✓ **Multi-Metric Forensics:** Laplacian stroke variance & edge density inspect physical printing sharpness alongside compression ELA.
- ✓ **Optical Disambiguation:** Heuristic OCR substitution rules (8↔6, O↔0) maintain mathematical checksum accuracy under glare.
- ✓ **Human-in-the-Loop Review:** System flags ambiguous inputs as SUSPICIOUS with clear diagnostic evidence for human officer decision-making.

Validated on 20-Document Controlled Testbed (10 Genuine + 10 Fake) | 98/98 Automated Tests Passed | Zero False Positives on Authentic Cards.

• Potential impact on the target audience:

- **Banks & NBFCs** → High-throughput automated customer KYC; eliminates loan fraud from forged salary slips, fake PAN, and fabricated Aadhaar cards.
- **Telecom Sector** → Real-time subscriber identity verification at point-of-sale for SIM allocation, shutting down synthetic identity rings.
- **Gig Platforms & Fleet Operators** → Instant, reliable driving licence and national ID verification for driver-partner and delivery agent onboarding.
- **Government Agencies & Welfare** → Fast citizen verification for direct benefit transfers, welfare distribution, and immigration transit checkpoints.
- **FinTech & Digital Onboarding** → Frictionless, privacy-preserving document fraud screening operating with sub-1.5s latency per document.

• Benefits of the solution (social, economic, environmental, etc.):

- ✓ **Economic Benefits:** Cuts manual screening overhead by >80%; prevents multi-crore fraud losses across financial institutions; sub-1.5s processing eliminates server scaling costs.
- ✓ **Social & Ethical Benefits:** Protects legitimate citizens against identity cloning; completely explainable verdict avoids algorithmic bias; strictly adheres to data privacy by retaining zero citizen PII.
- ✓ **Operational Efficiency:** Automated hierarchical evidence fusion reduces human reviewer fatigue by automatically verifying clean documents and highlighting exact tampering hotspots on suspicious cards.
- ✓ **Environmental & Resource Sustainability:** Lightweight, edge-deployable architecture requires minimal power and standard CPU hardware, avoiding carbon-heavy multi-billion parameter AI infrastructure.

- **Details / Links of the reference and research work:**

1. Verhoeff, J. (1969). Error Detecting Decimal Codes. Mathematical Centre Tracts 29.

– Foundational mathematical proof of the D5 dihedral group checksum algorithm used for Indian 12-digit Aadhaar UID verification with optical confusion tolerance.

2. Fischler, M. A., & Bolles, R. C. (1981). Random Sample Consensus: A paradigm for model fitting. Communications of the ACM, 24(6), 381–395.

– Core algorithm utilized in DocuShield's copy-move forgery detector to estimate geometric homography and eliminate false-positive keypoint clusters. DOI: 10.1145/358669.358692

3. Zhu, Y., Shen, C., & Zhao, H. (2015). Copy-move forgery detection based on scaled ORB. Multimedia Tools and Applications.

– Technique adapted for scale-invariant ORB keypoint extraction and spatial displacement clustering for digital document seal and signature verification. DOI: 10.1007/s11042-014-2431-2

4. Warif, N. B. A., Wahab, A. W. A., et al. (2015). An evaluation of Error Level Analysis in image forensics. IEEE International Conference on Systems, Engineering & Technology.

– Theoretical and empirical validation of 90% JPEG recompression error grids for identifying cut-and-paste tampering seams and font splicing. DOI: 10.1109/ICSEngT.2015.7412439

5. Sudiatmika, I. B. K., Rahman, M. A., et al. (2019). Image forgery detection using error level analysis and deep learning. TELKOMNIKA, 17(2), 653–659.

– Demonstrates the efficacy of localized ELA feature maps for document tampering detection and edge discrepancy classification. DOI: 10.12928/TELKOMNIKA.v17i2.8976

6. Vaidya, A., & Awasthi, A. (2025). Zero-to-One IDV: A Conceptual Model for AI-Powered Identity Verification. arXiv:2503.08734 [cs.CV].

– Architectural reference for modern confidence-aware identity document screening and multi-family forensic corroboration pipelines.